

# Power Electronics in Solar Industry to Soon Hit the Mainstream

With the ever-increasing demand for power, industry experts say the solar industry in India is still in its infancy. This story covers the scope for a career in the power electronics area of solar industry, including skills expected, pay packages and expert advice

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Each year our planet receives 5000 times more solar energy than it consumes. Using photovoltaic (PV) technology, it is possible to convert that energy into electricity, thereby helping to solve our present energy challenges.

Let us find out if there are enough job opportunities in the Indian solar industry for engineers with a power electronics background.

## Scope

The solar industry is growing in two ways. One is through the new solar power plants and the other is through solar-based products. The first vertical requires a person to be well versed in mechanical design, civil engineering, electromechanical engineering, electrical and, obviously, power electronics.

P. Chow Reddy, managing director, Interleaved Technologies says, "An engineer who has worked in these fields or similar ones can have an opportunity to give his services in the field. But for the second vertical, one having basic power electronics knowledge and experience can only make an entry."

The scope for a career in power electronics in solar field is immense. "Indians are really good at inverters and UPS. Solar adds a few more skills and expertise into the mix. Anyone who is good at these three or four years from now will be in huge demand, especially when solar hits the mainstream," says Deyan Alex, director of operations, PowerIt Renewable Energy Pvt Ltd.

"There is an acute shortage of skilled manpower in functional areas

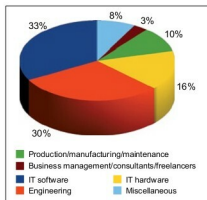


Fig. 1: Top functional areas (Data courtesy: TimesJobs.com)

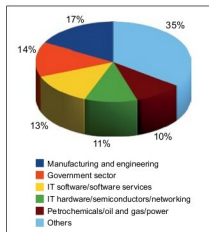


Fig. 2: Top industries hiring power electronics engineers (Data courtesy: TimesJobs.com)

such as area site survey, design and engineering, execution, installation and commissioning, operation and maintenance," informs Chetan Kumar, assistant director, Paawan Energy India Pvt Ltd. For solar PV (SPV) industry, the most preferred backgrounds are B. Tech. in EC/EE/EI/ME.

"Considering the components that go into a solar energy system, after energy storage (that is, batteries), the next most sought-after components are the power electronics circuits required for energy conversion," informs Ganesh Shankar, co-founder and managing director, FluxGen Engineering Technologies.

"Engineers with strong hold in circuit theory and solid-state electronics are much required to particularly address the reliability and efficiency aspects of power electronics in the solar industry," says Shankar.

Power electronics is a lucrative field that has application in both smart grid regulation technologies as well as solid-state inverter electronics for the control and conversion of electric

power. Energy companies, battery and genset manufacturers, electronics and computer hardware manufacturers, and power generation, conversion and distribution companies look for people with these skills. "Solar energy electronics is still a nascent field and has a great potential for growth in the near future. Currently, it is a highly niche segment, with most people having risen to positions with a strong background in allied power-generation technologies," informs Vivek Madhukar, COO, Times Business Solutions Limited (which operates TimesJobs.com).

The major share of the jobs available for engineers with a power electronics background is in functional areas such as manufacturing and engineering which constitutes about 17 per cent, followed by IT software sector which provides about 13 per cent and the semiconductor sector with about 11 per cent of the jobs. Around 14 per cent of the jobs are provided by the government.